

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Nov 2025

Course Code: EEE101R01

Course: BASIC ELECTRICAL ENGINEERING

QP No. :S1701-1

Duration: 3 hours

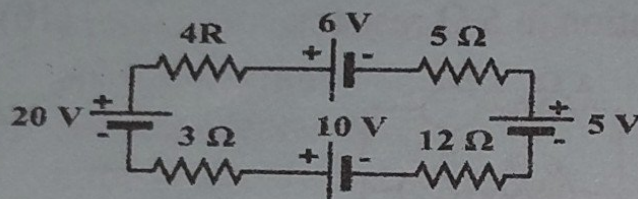
Max. Marks:100

PART - A

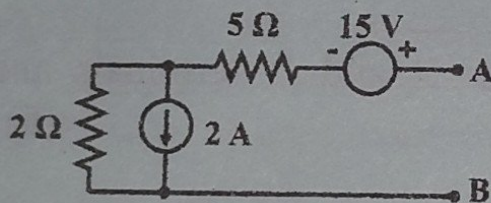
Answer all the questions

10 x 2 = 20 Marks

1. Determine the value of R such that the power dissipated in the $5\ \Omega$ resistor is 1.25 W .



2. Transform the given circuit into a single current source equivalent across terminals AB.



3. A voltage of $v(t) = 50\sin(314t + 95^\circ)$ volts applied to a two-element series circuit results in a current of $i(t) = 5\sin(314t + 65^\circ)$. Find the impedance and identify the elements connected in the circuit.
4. A parallel circuit having an equivalent admittance of $Y = 0.09 \angle -56.3^\circ$ mho is connected across a 220 V, 50 Hz ac supply. Obtain the complex power S .
5. A sinusoidal alternating current of frequency 25 Hz has a maximum value of 100 A. How long will it take to reach an instantaneous value of 20 A.
6. Two identical coils A and B consisting of 1200 turns each lie in parallel planes such that 80% of the flux produced by the current in coil A link the coil B. A current of 5 A flowing in the coil A produces in it a flux of 0.04 mWb. Calculate the mutual inductance.

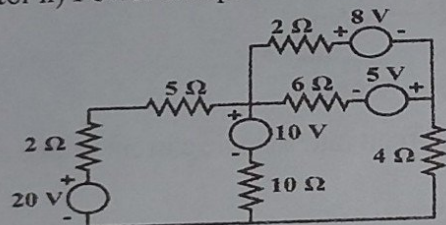
7. A coil of 400 turns is wound over an iron ring with a relative permeability of 1300. The mean circumference of the ring is 50 cm and the cross-sectional area is 12 cm^2 . Calculate the reluctance.
8. Justify the significance of starter in a DC motor.
9. Three similar coils each of impedance $(10+j5) \Omega$ are connected in star across a 3-phase 440 V, 50 Hz supply. Find the line and phase values of current.
10. Mention any two IE rules related to domestic wiring.

PART - B

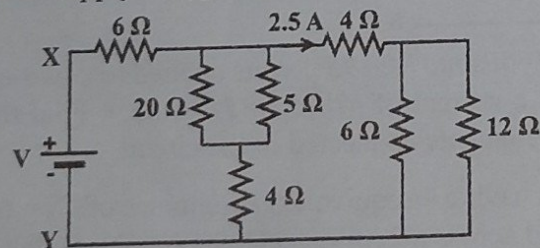
4 x 15 = 60 Marks

Answer any Four questions

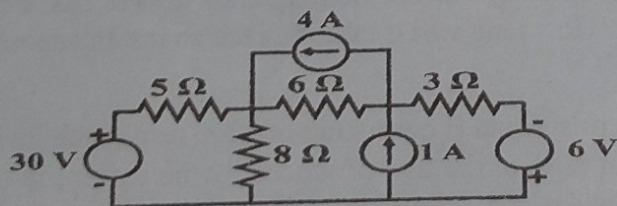
11. a) Apply mesh analysis for the given circuit and solve for the i) Current in 5Ω resistor ii) Power dissipation in 6Ω resistor. (10)



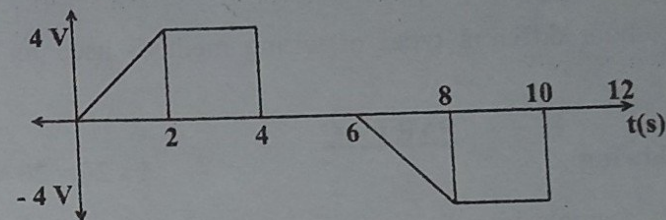
- b) Determine Equivalent resistance across X and Y terminals and calculate the supply voltage V. (5)



12. a) Apply Nodal analysis for the following circuit and determine i) Current delivered by 30 V source ii) Voltage across 6Ω resistor. (10)



- b) Calculate R.M.S value and average value of the following waveform. (5)



13. a) A circuit of resistance 50Ω and inductance of 80 mH is connected in series with a capacitor. When this combination is connected across an ac supply of 200 V, 50 Hz, the current drawn by the circuit is 3.8 A. Determine i) Impedance ii) Capacitance iii) power factor iv) expression of instantaneous voltage and current. Also draw the phasor diagram. (10)
- b) An impedance of $(15 - j15) \Omega$ is connected in parallel with an impedance of $(5 + j10) \Omega$. If the supply voltage is 200 V, 50 Hz, Determine i) equivalent admittance ii) branch currents and total current. (5)
14. a) Describe the construction of a DC machine with neat diagram and explain the function of each part of the machine. (10)
- b) Derive the expression of induced emf in transformer. (5)
15. a) How the DC motors are classified based on the field winding connection with armature winding? Illustrate each type with appropriate equivalent circuit. (8)
- b) A school has the following electrical installations connected to a 230 V supply. Find the weekly energy consumption for a 6-day week. (7)

Load	Power rating (W)	Daily usage in hours
Lighting	200 lamps of 100 W each and 100 lamps of 60 W each	6
Fans	150 fans of 75 W each	8
Computers	50 systems of 250 W each	7
Air-conditioners	20 units of 1.5 kW each.	5

16. a) State the necessity of earthing in electrical systems. How is pipe earthing carried out in residential buildings? Support your answer with a neat sketch. (8)
- b) Explain about different types of wiring methods used for house wiring. (7)

PART - C

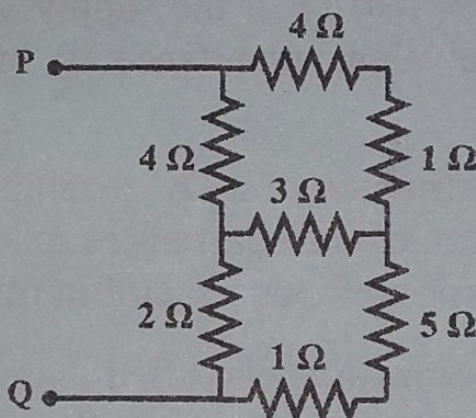
Answer the following

1 x 20 = 20 Marks

17. a) Construct a single line diagram of electric power supply scheme including various stages of transmission and distribution with appropriate change in voltage levels starting from generating station to consumer end. (5)

In the consumer side of the above power supply system, the following loads as described in sections (b), (c) and (d) are connected to the single phase 230 V, 50 Hz ac supply.

- b) Determine equivalent resistance of load 1 connected between P and Q terminals. (5)



- c) Load 2 with an impedance of $Z_L = (15 + j10)\Omega$ is connected across the given 230 V ac supply. Find the supply current and power factor. If a capacitance of $X_c = -j 32.5\Omega$ is connected in parallel with Z_L , calculate the new supply current and power factor for this combination. Also write your inference with reference to the phasor diagram. (6)
- d) Load 3 is connected to the supply through a step down transformer with the following ratings: 2 kVA, 230/110 V, 50 Hz, primary turns 460, maximum flux density in the core is 1.1 Tesla. Calculate i) transformation ratio ii) secondary turns iii) cross-sectional area of core. (4)
